AGRIVIVE / SABAI-ROOT

ODISHA / SABAI-CRAFT INSPIRED / EROSION-MAT ANCHOR

SERIES 19 / 28

A3 / PRELIMINARY / ALL DIMENSIONS mm



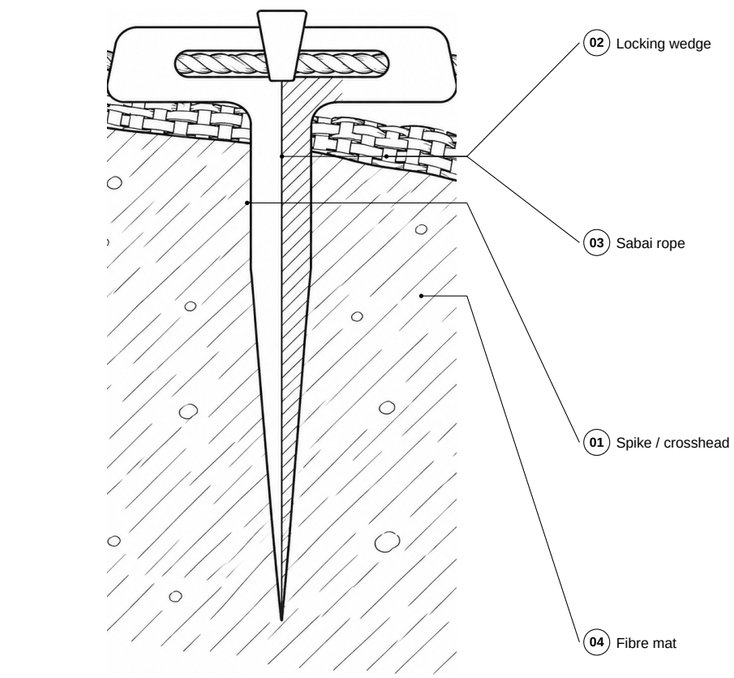
STATE CONTEXT / INDIGENOUS INSPIRATION

Sabai fibre craft in Odisha includes ropes and mats. SABAI-ROOT proposes a slotted soil anchor and locking wedge for a fibre mat; this new application has no established pull-out rating.

Practice source: Mayurbhanj district: Sabai grass products | Outline: geoBoundaries / schematic

01 INSTALLED SOIL / WATER SECTION

NOMINAL ENVELOPE / mm: 100 crosshead span x 160 spike L

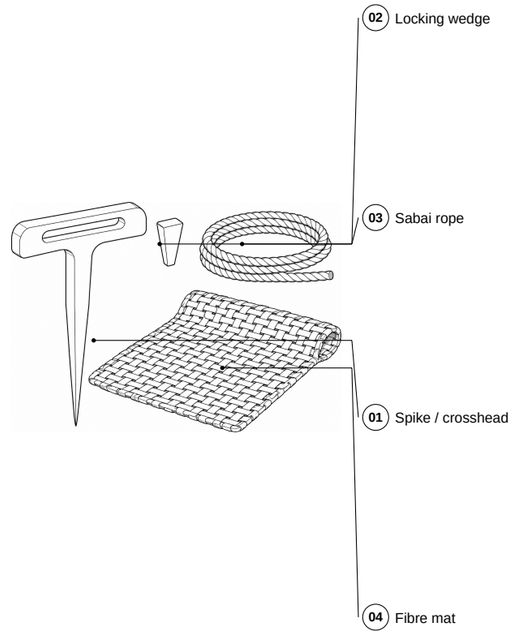


DESIGN / OPERATION NOTES

Lay a suitable fibre mat over prepared soil.
Drive the spike through the mat with the crosshead above the surface.
Thread the rope through the slot and seat the removable wedge.

02 EXPLODED ASSEMBLY

ASSEMBLED ENVELOPE / mm
100 crosshead span x 160 spike L

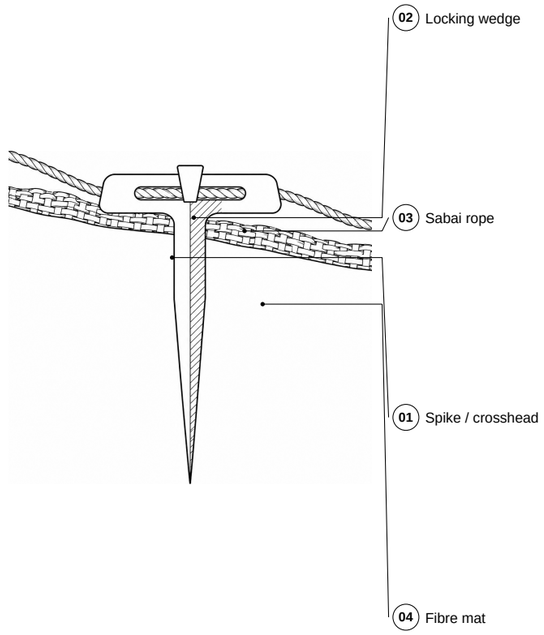


DESIGN / OPERATION NOTES

Components separated for service and assembly.
Connections and mating fits require CAD.

04 INTERNAL SECTION

SECTION BASIS / mm
8 nominal spine thick; mat-retention slot 16 x 5



DESIGN / OPERATION NOTES

Drive the spike through the mat with the crosshead above the surface.
Thread the rope through the slot and seat the removable wedge.

DESIGN / INTERFACE SCHEDULE

01 NOMINAL GEOMETRY

Crosshead span	**100**
Spike length / thickness	**160 / 8**
Retention slot W x H	**16 x 5**

02 GEOMETRIC CHECK

RETENTION SLOT AREA

80.0 mm2

16 x 5
Pull-out capacity depends on soil and mat construction

03 PARTS / QTY

01 Spike / crosshead	**1**
02 Locking wedge	**1**
03 Sabai rope	**As needed**
04 Fibre mat	**Site item**

Spike/wedge: candidate polymer. Rope/mat: natural Sabai fibre; construction and grade TBD.

04 INTERFACES / CHECKS

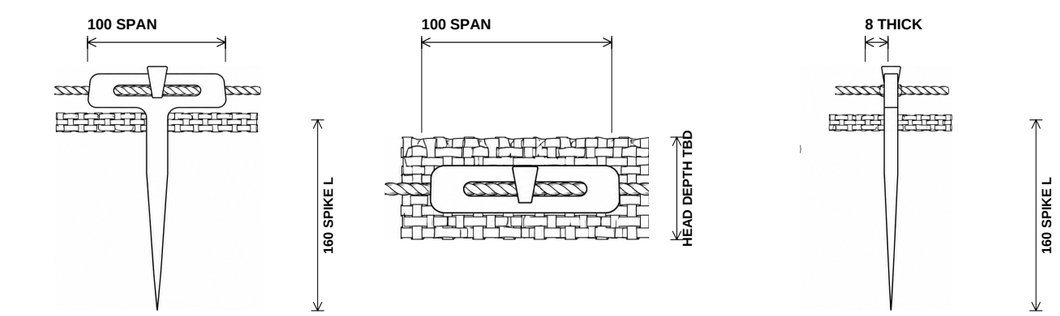
Test pull-out and crosshead strength in actual soil.
Check rope abrasion and wedge retention.
Assess mat spacing and runoff scour independently.

05 DATUMS / DRAWING BASIS

A: crosshead bearing face. B: spike axis. Keep rope clear of sharp slot edges.

PRELIMINARY / VALIDATION PENDING

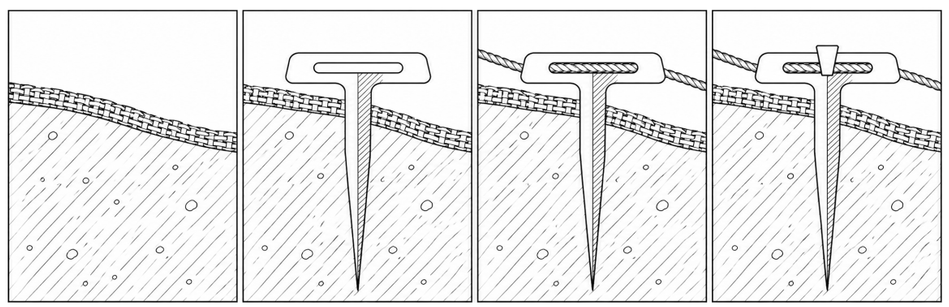
03 ORTHOGRAPHIC STUDIES / mm



FRONT / CROSSHEAD **TOP / CROSSHEAD** **SIDE / SPINE**

Crosshead 100 span; soil spike 160 L, nominal spine 8 thick; retention slot 16 W x 5 H.

05 INSTALLATION / OPERATING SEQUENCE



1 LAY MAT **2 SET SPIKE** **3 THREAD ROPE** **4 FIT WEDGE**

Concept sequence; validate fit, access and operating conditions in prototypes.

DRAWING / REVISION	TITLE	STATUS	REVISION DATE	FORMAT / SCALE / UNITS	COLLECTION SHEET
AGR-19 / P01	SABAI-ROOT	PRELIMINARY CONCEPT / CAD & TRIALS PENDING	10 NOV 2026	A3 / NTS / mm	19 / 28